# An Introduction to Relative Langlands Duality

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November 2026

**Abstract**

These are notes for an introductory lecture on the relative Langlands duality proposed by Ben-Zvi, Sakellaridis and Venkatesh [BZSV24]. After recalling the structure theory of algebraic groups needed to define the Langlands dual group, we explain how periods of automorphic forms are indexed by spherical varieties and, more intrinsically, by hyperspherical Hamiltonian spaces; we state the local, global and numerical forms of the duality; and we describe the four-dimensional topological field theory picture that organizes them. The last section explains how Hamiltonian spaces appear as Lagrangians in shifted symplectic stacks, and lists some questions. No originality is claimed.

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# 1 Introduction

Automorphic forms carry two kinds of numerical invariants. Integrating an automorphic form against a distinguished cycle gives a *period*; evaluating the $L$-function of its Langlands parameter gives an *$L$-value*. Since Hecke's work in 1937, many identities of the form

$$\text{period of } \varphi \;\doteq\; \text{special value of an } L\text{-function of } \varphi$$

have been found, where $\doteq$ means equality up to explicit and well-understood factors. In the language of the Langlands program these identities say that a diagram

$$\begin{array}{ccc} \{\text{automorphic forms on } G\} & \overset{\text{Langlands}}{\longleftrightarrow} & \{\text{Galois representations into } G^\vee\} \\ & \searrow^{\text{period}} \quad \swarrow_{L\text{-function}} & \\ & \mathbb{C} & \end{array} \tag{1.1}$$

commutes. Relative Langlands duality, as formulated by Ben-Zvi, Sakellaridis and Venkatesh [BZSV24], proposes a systematic explanation:

1. periods for $G$ are indexed by Hamiltonian $G$-spaces $M$;
2. $L$-functions for $G^\vee$ are indexed by Hamiltonian $G^\vee$-spaces $M^\vee$;
3. there is a duality $(G, M) \leftrightarrow (G^\vee, M^\vee)$ on a well-behaved class of such spaces, the *hyperspherical* ones, under which (1.1) commutes;
4. analogous statements hold in every "tier" of the Langlands program: local and global, arithmetic and geometric.

A first example, from the introduction of [BZSV24]: for $G = \mathrm{GL}_n \times \mathrm{GL}_n$, the Godement–Jacquet and Rankin–Selberg integrals are dual to each other. The Godement–Jacquet period of $\varphi$ is the $L$-function indexed by the Rankin–Selberg space, and vice versa. Classically these are two separate theorems [GJ72, JPSS83]; from the present point of view they are one statement read in two directions.

**Outline.** Section 2 recalls the structure theory of linear algebraic groups up to reductive groups. Section 3 defines root data and the Langlands dual group. Section 4 discusses periods and spherical varieties, and Section 5 hyperspherical varieties and the duality. Section 6 states the conjectures, Section 7 explains the field-theoretic picture, and Section 8 discusses shifted symplectic geometry and open questions. The report [BZSV25] of the 2025 Oberwolfach Arbeitsgemeinschaft is a useful companion survey.

**Conventions.** In Sections 2–3, $k$ is an algebraically closed field and an *algebraic group* means a smooth affine group scheme of finite type over $k$; algebraic subgroups are closed. From Section 4 on, $G$ is a split connected reductive group over a field $F$, and the dual group $G^\vee$ is defined over a coefficient field of characteristic zero ($\mathbb{C}$ or $\overline{\mathbb{Q}}_\ell$). We work in the unramified setting throughout. When $F$ is a local field we write $\mathcal{O}$ for its ring of integers, typically $F = \mathbb{C}((t))$, $\mathcal{O} = \mathbb{C}[[t]]$ in the geometric setting. Page numbers for [BZSV24] refer to the version on Sakellaridis's webpage; section numbers agree with the arXiv version.

**Acknowledgements.** These notes were prepared with the help of Claude, an AI model developed by Anthropic, which was used to draft the text, examples and references. The author has reviewed the notes, but readers should check statements against the original sources cited.

# 2 Background on algebraic groups

The goal of this section is the definition of a reductive group. The definition says that a certain normal subgroup, the unipotent radical, is trivial; to make sense of it we need solvable, nilpotent and unipotent groups. Standard references are [Mil17, Ch. 6, 14, 16, 17, 19], [Spr98] and [Hum75].

## 2.1 Examples

**Example 2.1.1.** The following algebraic groups will be used throughout.

- $\mathbb{G}_a = (k, +)$, the additive group, with coordinate ring $k[x]$.
- $\mathbb{G}_m = (k^\times, \cdot)$, the multiplicative group, with coordinate ring $k[x, x^{-1}]$.
- A *torus* is an algebraic group isomorphic to $\mathbb{G}_m^r$ for some $r$.
- $\mathrm{GL}_n$, $\mathrm{SL}_n$, $\mathrm{PGL}_n = \mathrm{GL}_n/\mathbb{G}_m$, $\mathrm{Sp}_{2n}$, $\mathrm{SO}_n$ (when orthogonal groups appear we assume $\mathrm{char}\, k \neq 2$).
- $B_n \subset \mathrm{GL}_n$, the upper triangular invertible matrices.
- $U_n \subset B_n$, the upper triangular matrices with 1's on the diagonal.
- $T_n \subset B_n$, the diagonal matrices; $T_n \cong \mathbb{G}_m^n$.

## 2.2 Subnormal series

**Definition 2.2.1.** A *subnormal series* of an algebraic group $G$ is a chain of algebraic subgroups

$$G = G_0 \supset G_1 \supset \cdots \supset G_s = \{e\}$$

such that each $G_{i+1}$ is normal in $G_i$. The groups $G_i/G_{i+1}$ are its *quotients*. A subnormal series is *normal* if each $G_i$ is normal in $G$, and *central* if moreover $G_i/G_{i+1}$ lies in the centre of $G/G_{i+1}$.

Quotients of algebraic groups by normal algebraic subgroups exist and are again algebraic groups [Mil17, Ch. 5], so this definition behaves as in abstract group theory.

**Example 2.2.2.** In $B_n$, let $U_n^{(j)}$ be the subgroup of $U_n$ of matrices $u$ with $u_{ab} = 0$ for $0 < b - a < j$, i.e. vanishing on the first $j-1$ superdiagonals. Then

$$B_n \supset U_n = U_n^{(1)} \supset U_n^{(2)} \supset \cdots \supset U_n^{(n)} = \{e\}$$

is a normal series. Its first quotient is $B_n/U_n \cong T_n \cong \mathbb{G}_m^n$ (take the diagonal), and each later quotient $U_n^{(j)}/U_n^{(j+1)}$ is isomorphic to $\mathbb{G}_a^{n-j}$ (take the $j$-th superdiagonal).

**Definition 2.2.3.** Let $N$ and $H$ be algebraic groups and let $H$ act on $N$ by group automorphisms, $h \mapsto (n \mapsto {}^h n)$. The *semidirect product* $H \ltimes N$ (also written $N \rtimes H$) is the variety $H \times N$ with multiplication

$$(h, n) \cdot (h', n') = \big(hh',\ {}^{h'^{-1}}n \cdot n'\big).$$

An algebraic group $G$ is the (internal) semidirect product $H \ltimes N$ of algebraic subgroups $H, N \subset G$ if $N$ is normal, $H \cap N = \{e\}$, and the multiplication map $H \times N \to G$, $(h, n) \mapsto hn$, is an isomorphism of varieties. Then $H$ acts on $N$ by conjugation and $G/N \cong H$.

The open side of the symbol faces the normal factor and the vertical bar is next to the acting group, so $H \ltimes N$ and $N \rtimes H$ denote the same group, with $N$ normal. For example, $B_n = T_n \ltimes U_n$: every invertible upper triangular matrix factors uniquely as a diagonal matrix times a unipotent upper triangular one,

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} 1 & b/a \\ 0 & 1 \end{pmatrix},$$

and $U_n$ is normal in $B_n$ while $T_n$ is not.

## 2.3 Solvable and nilpotent groups

For an algebraic group $G$, the *derived group* $\mathcal{D}G = [G, G]$ is the smallest algebraic subgroup containing all commutators; it is normal, and connected if $G$ is. Set $\mathcal{D}^0 G = G$ and $\mathcal{D}^{i+1} G = \mathcal{D}(\mathcal{D}^i G)$ (derived series), and $C^0 G = G$, $C^{i+1} G = [G, C^i G]$ (lower central series).

**Definition 2.3.1.** An algebraic group $G$ is

1. *solvable* if it has a subnormal series with commutative quotients; equivalently, $\mathcal{D}^s G = \{e\}$ for some $s$;
2. *nilpotent* if it has a central series; equivalently, $C^s G = \{e\}$ for some $s$.

Nilpotent groups are solvable. Subgroups, quotients and extensions of solvable groups are solvable; subgroups and quotients of nilpotent groups are nilpotent.

**Example 2.3.2.**

1. Tori and $\mathbb{G}_a$ are commutative, hence nilpotent.
2. $B_n$ is solvable by Example 2.2.2.
3. $U_n$ is nilpotent: the series $U_n^{(j)}$ is central.
4. $B_2$ is solvable but not nilpotent. Its centre is the scalar matrices $Z$, and $B_2/Z \cong \mathbb{G}_a \rtimes \mathbb{G}_m$, with $\mathbb{G}_m$ acting on $\mathbb{G}_a$ by scaling, has trivial centre. A nontrivial nilpotent group has nontrivial centre and quotients of nilpotent groups are nilpotent, so $B_2/Z$, and hence $B_2$, is not nilpotent.
5. $\mathrm{GL}_n$ and $\mathrm{SL}_n$ ($n \geq 2$) are not solvable: $\mathcal{D}\mathrm{SL}_n = \mathrm{SL}_n$.

**Theorem 2.3.3** (Lie–Kolchin)**.** *Let $G \subset \mathrm{GL}(V)$ be a connected solvable algebraic group. Then $G$ fixes a full flag in $V$. Equivalently, in a suitable basis $G \subset B_n$.*

So $B_n$ is, up to conjugation, the universal connected solvable subgroup of $\mathrm{GL}_n$. This motivates:

**Definition 2.3.4.** A *Borel subgroup* of $G$ is a maximal connected solvable algebraic subgroup.

**Theorem 2.3.5** (Borel)**.** *Let $G$ be connected. Any two Borel subgroups of $G$ are conjugate, $G/B$ is a projective variety, and any two maximal tori of $G$ are conjugate. Every maximal torus lies in some Borel subgroup.*

For $G = \mathrm{GL}_n$, the Borel subgroups are the conjugates of $B_n$, and $\mathrm{GL}_n/B_n$ is the variety of full flags in $k^n$.

**Definition 2.3.6.** Let $G$ be connected and $T \subset G$ a maximal torus, with normalizer $N_G(T) = \{g \in G : gTg^{-1} = T\}$ and centralizer $Z_G(T)$. The *Weyl group* of $(G, T)$ is

$$W = W(G, T) = N_G(T)/Z_G(T).$$

Since $N_G(T)$ normalizes $T$, the group $W$ acts on $T$ by conjugation, and hence on the lattices $X^*(T)$ and $X_*(T)$ by $(w\chi)(t) = \chi(w^{-1}tw)$ and $(w\lambda)(s) = w\lambda(s)w^{-1}$. The Weyl group is finite, and by Theorem 2.3.5 it does not depend on the choice of $T$ up to isomorphism. If $G$ is reductive then $Z_G(T) = T$, so $W = N_G(T)/T$.

**Example 2.3.7.**

1. For $G = \mathrm{GL}_n$ and $T = T_n$, the normalizer $N_G(T)$ consists of the *monomial matrices*, with exactly one nonzero entry in each row and column. Every monomial matrix is a permutation matrix times a diagonal matrix, so
$$W \cong S_n,$$
acting on $T_n$ by permuting the diagonal entries:
$$\sigma \cdot \mathrm{diag}(t_1, \ldots, t_n) = \mathrm{diag}(t_{\sigma^{-1}(1)}, \ldots, t_{\sigma^{-1}(n)}).$$
2. For $G = \mathrm{SL}_2$, $W = \{1, w\} \cong \mathbb{Z}/2$ with $w$ represented by $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, acting on $T$ by $\mathrm{diag}(s, s^{-1}) \mapsto \mathrm{diag}(s^{-1}, s)$.
3. For a torus $G = T$, $W$ is trivial.

**Exercise 2.3.8.** Show that conjugation by $N_G(T)$ permutes the eigenspace decomposition of $k^n$ under $T_n$ in Example 2.3.7(1), and deduce that every element of $N_{\mathrm{GL}_n}(T_n)$ is monomial.

## 2.4 Unipotent groups

**Definition 2.4.1.** An element $g \in \mathrm{GL}(V)$ is *unipotent* if $g - 1$ is nilpotent. An algebraic group $U$ is *unipotent* if every nonzero representation $V$ of $U$ has a nonzero fixed vector.

Iterating the defining property, every finite-dimensional representation of a unipotent group has a full flag of invariant subspaces on which $U$ acts trivially on the quotients. Hence:

**Proposition 2.4.2.** *An algebraic group is unipotent if and only if it is isomorphic to an algebraic subgroup of $U_n$ for some $n$, if and only if all its elements are unipotent. Unipotent groups are nilpotent.*

**Example 2.4.3.** $\mathbb{G}_a$ and $U_n$ are unipotent. A torus is never unipotent unless trivial: $\mathbb{G}_m$ acts on $k$ by the identity character with no fixed vector. Unipotent groups and tori are the two opposite extremes; by the Jordan decomposition, every element of $\mathrm{GL}_n$ factors uniquely as $g = g_s g_u = g_u g_s$ with $g_s$ semisimple and $g_u$ unipotent.

**Theorem 2.4.4** (Structure of connected solvable groups)**.** *Let $G$ be connected and solvable, and let $G_u$ be the set of unipotent elements. Then $G_u$ is a connected normal algebraic subgroup, and for any maximal torus $T \subset G$,*

$$G = T \ltimes G_u.$$

For $G = B_n$ this is the decomposition $B_n = T_n \ltimes U_n$.

## 2.5 The radical and the unipotent radical

Let $G$ be connected. If $N_1, N_2 \subset G$ are connected normal solvable subgroups, so is $N_1N_2$, being an extension of $N_2/(N_1 \cap N_2)$ by $N_1$. Hence there is a largest one, and similarly for unipotent subgroups.

**Definition 2.5.1.** Let $G$ be a connected algebraic group.

1. The *radical* $R(G)$ is the largest connected normal solvable algebraic subgroup of $G$.
2. The *unipotent radical* $R_u(G)$ is the largest connected normal unipotent algebraic subgroup of $G$.

Since unipotent groups are solvable, $R_u(G) \subset R(G)$; in fact $R_u(G) = R(G)_u$ by Theorem 2.4.4.

**Example 2.5.2.**

1. $R(\mathrm{GL}_n) = Z(\mathrm{GL}_n) = \mathbb{G}_m$ (scalar matrices) and $R_u(\mathrm{GL}_n) = \{e\}$.
2. $R(\mathrm{SL}_n) = R_u(\mathrm{SL}_n) = \{e\}$.
3. $R(B_n) = B_n$ and $R_u(B_n) = U_n$.
4. Let $P \subset \mathrm{GL}_n$ be the block upper triangular matrices with diagonal blocks of sizes $n_1, \dots, n_r$ (a *parabolic* subgroup). Then $R_u(P)$ is the subgroup of block upper triangular matrices whose diagonal blocks are identity matrices, and

$$P = L \ltimes R_u(P), \qquad L = \mathrm{GL}_{n_1} \times \cdots \times \mathrm{GL}_{n_r} \text{ (the Levi factor).}$$

## 2.6 Reductive and semisimple groups

**Definition 2.6.1.** A connected algebraic group $G$ is

1. *reductive* if $R_u(G) = \{e\}$;
2. *semisimple* if $R(G) = \{e\}$.

Over a field that is not algebraically closed, one requires these conditions after base change to an algebraic closure.

Semisimple groups are reductive. By Example 2.5.2, $\mathrm{GL}_n$ is reductive but not semisimple, $\mathrm{SL}_n$ is semisimple, and $B_n$ and proper parabolic subgroups are not reductive. The groups $\mathrm{PGL}_n$, $\mathrm{Sp}_{2n}$, $\mathrm{SO}_n$ ($n \geq 3$) are semisimple, and tori are reductive.

**Theorem 2.6.2.** *Let $G$ be a connected algebraic group.*

1. *If $G$ is reductive, then $R(G) = Z(G)^\circ$ is a torus, $\mathcal{D}G$ is semisimple, and the multiplication map $R(G) \times \mathcal{D}G \to G$ is a surjection with finite kernel. For instance $\mathrm{GL}_n = \mathbb{G}_m \cdot \mathrm{SL}_n$ with kernel $\mu_n$.*
2. *If $\operatorname{char} k = 0$, $G$ is reductive if and only if every finite-dimensional representation of $G$ is completely reducible.*
3. *(Levi decomposition, $\operatorname{char} k = 0$.) There is a reductive subgroup $L \subset G$ with $G = L \ltimes R_u(G)$.*

Part (2) explains the name: $\mathbb{G}_a$ acting on $k^2$ by $x \mapsto \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$ preserves the line $ke_1$ but has no invariant complement. Part (3) says that every connected group is built from a reductive group and a unipotent group, which is why reductive groups are the central objects of the theory.

**Exercise 2.6.3.** Show directly that $R_u(\mathrm{GL}_n) = \{e\}$: if $N \subset \mathrm{GL}_n$ is a normal unipotent subgroup, the subspace $(k^n)^N$ of $N$-fixed vectors is nonzero and $\mathrm{GL}_n$-stable, hence equal to $k^n$.

## 2.7 The Lie algebra of an algebraic group

The Lie algebra is the linearization of $G$ at the identity. It will be used to define roots in Section 3, and its dual $\mathfrak{g}^*$ is the target of moment maps in Section 5. References are [Mil17, Ch. 10], [Hum75, §9–10] and [Spr98, Ch. 4].

*Notation.* We denote the Lie algebra of a group by the corresponding fraktur letter: $\mathfrak{g} = \mathrm{Lie}(G)$, $\mathfrak{h} = \mathrm{Lie}(H)$ for a subgroup $H$, $\mathfrak{t} = \mathrm{Lie}(T)$ for a torus $T$, and so on. Many texts write $\mathfrak{h}$ for a Cartan subalgebra; in these notes Cartan subalgebras are always denoted $\mathfrak{t}$, and $\mathfrak{h}$ is always the Lie algebra of a subgroup $H$.

Let $k[\varepsilon] = k[x]/(x^2)$ be the ring of dual numbers. Since $G$ is a functor on $k$-algebras, the map $k[\varepsilon] \to k$, $\varepsilon \mapsto 0$, induces a group homomorphism $G(k[\varepsilon]) \to G(k)$.

**Definition 2.7.1.** The *Lie algebra* of $G$ is

$$\mathfrak{g} = \mathrm{Lie}(G) = \ker\big(G(k[\varepsilon]) \to G(k)\big).$$

Equivalently, $\mathfrak{g}$ is the tangent space $T_eG$ of $G$ at the identity; since $G$ is smooth, $\dim_k \mathfrak{g} = \dim G$.

For $G \subset \mathrm{GL}_n$, an element of $\mathfrak{g}$ is a matrix $1 + \varepsilon X$ with $X \in \mathfrak{gl}_n = M_n(k)$ satisfying the equations of $G$ modulo $\varepsilon^2$; we identify it with $X$.

**Example 2.7.2.**

1. $\mathrm{Lie}(\mathrm{GL}_n) = \mathfrak{gl}_n = M_n(k)$, since $1 + \varepsilon X$ is invertible with inverse $1 - \varepsilon X$.
2. $\mathrm{Lie}(\mathrm{SL}_n) = \mathfrak{sl}_n = \{X : \operatorname{tr} X = 0\}$, since $\det(1 + \varepsilon X) = 1 + \varepsilon \operatorname{tr} X$.
3. $\mathrm{Lie}(\mathbb{G}_a) = \mathrm{Lie}(\mathbb{G}_m) = k$, and $\mathrm{Lie}(T_n)$ is the diagonal matrices.
4. $\mathrm{Lie}(B_n) = \mathfrak{b}_n$ and $\mathrm{Lie}(U_n) = \mathfrak{n}_n$, the upper triangular and strictly upper triangular matrices.
5. If $G = \{g \in \mathrm{GL}_n : g^T J g = J\}$ for a nondegenerate form $J$ (e.g. $\mathrm{Sp}_{2n}$ or $\mathrm{SO}_n$), then $\mathfrak{g} = \{X : X^T J + JX = 0\}$.

**Exercise 2.7.3.** Verify (2) and (5) of Example 2.7.2, and check that $\dim \mathfrak{sl}_n = n^2 - 1 = \dim \mathrm{SL}_n$.

**The adjoint representation and the bracket.** $G$ acts on itself by conjugation, $c_g(x) = gxg^{-1}$, fixing $e$. Differentiating at $e$ gives a representation

$$\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g}), \qquad \mathrm{Ad}(g)X = gXg^{-1} \ \text{ for } G \subset \mathrm{GL}_n,$$

the *adjoint representation*. Its differential $\mathrm{ad} = d\,\mathrm{Ad} : \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ defines the *Lie bracket* $[X, Y] = \mathrm{ad}(X)(Y)$. For $G \subset \mathrm{GL}_n$ this is the commutator $[X, Y] = XY - YX$. The bracket is bilinear, alternating and satisfies the Jacobi identity, so $\mathfrak{g}$ is a Lie algebra.

**The coadjoint representation.** The dual vector space $\mathfrak{g}^* = \mathrm{Hom}_k(\mathfrak{g}, k)$ carries the dual of the adjoint action,

$$\mathrm{Ad}^*(g)\lambda = \lambda \circ \mathrm{Ad}(g^{-1}).$$

For $G = \mathrm{GL}_n$, the trace form $(X, Y) \mapsto \operatorname{tr}(XY)$ is nondegenerate and Ad-invariant, so $\mathfrak{gl}_n^* \cong \mathfrak{gl}_n$ as representations; similarly for any reductive $G$ in characteristic 0 via a nondegenerate invariant form.

**Functoriality.** A homomorphism $\varphi : G \to H$ induces a Lie algebra homomorphism $d\varphi = \mathrm{Lie}(\varphi) : \mathfrak{g} \to \mathfrak{h}$. A representation $\rho : G \to \mathrm{GL}(V)$ thus gives a representation $d\rho : \mathfrak{g} \to \mathfrak{gl}(V)$.

**Example 2.7.4** ($\mathfrak{sl}_2$)**.** The Lie algebra $\mathfrak{sl}_2$ has basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

with $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. The diagonal torus $t = \mathrm{diag}(s, s^{-1})$ acts by $\mathrm{Ad}(t)e = s^2 e$, $\mathrm{Ad}(t)f = s^{-2} f$, $\mathrm{Ad}(t)h = h$. Thus the weights of the torus on $\mathfrak{sl}_2$ are 0 and $\pm 2$; these are the roots of $\mathrm{SL}_2$ in Example 3.2.3. A triple $(e, h, f)$ in a Lie algebra $\mathfrak{g}$ with these relations is an $\mathfrak{sl}_2$-*triple*; in characteristic 0 such triples correspond to homomorphisms $\mathrm{SL}_2 \to G$, and they appear in the structure theorem for hyperspherical varieties (Theorem 5.4.1).

**Cartan subalgebras.** In Section 3 the roots are defined by decomposing $\mathfrak{g}$ under a maximal torus $T$; the piece on which $T$ acts trivially is $\mathrm{Lie}(T)$. This subalgebra has an intrinsic description.

**Definition 2.7.5.** Let $\mathfrak{g}$ be a finite-dimensional Lie algebra over $k$. A *Cartan subalgebra* of $\mathfrak{g}$ is a subalgebra $\mathfrak{t} \subset \mathfrak{g}$ that is nilpotent and self-normalizing, i.e. $\{X \in \mathfrak{g} : [X, \mathfrak{t}] \subset \mathfrak{t}\} = \mathfrak{t}$.

**Theorem 2.7.6.** *Let* $\operatorname{char} k = 0$ *and let* $G$ *be connected reductive with Lie algebra* $\mathfrak{g}$*.*

1. *The Cartan subalgebras of* $\mathfrak{g}$ *are exactly the subalgebras* $\mathrm{Lie}(T)$ *for* $T \subset G$ *a maximal torus. They are abelian and consist of semisimple elements; equivalently, they are the maximal subalgebras of* $\mathfrak{g}$ *consisting of semisimple elements (maximal toral subalgebras).*
2. *Any two Cartan subalgebras are conjugate under* $\mathrm{Ad}(G)$*, by Theorem 2.3.5. Their common dimension is the* rank *of* $G$*.*
3. *(Chevalley restriction theorem.) Restriction of functions gives an isomorphism* $k[\mathfrak{g}]^G \cong k[\mathfrak{t}]^W$*, where* $W$ *is the Weyl group. Thus* $\mathfrak{g} /\!\!/ G \cong \mathfrak{t}/W$*.*

**Example 2.7.7.**

1. In $\mathfrak{gl}_n$, the diagonal matrices $\mathfrak{t}_n = \mathrm{Lie}(T_n)$ form a Cartan subalgebra, and $\mathrm{GL}_n$ has rank $n$. In $\mathfrak{sl}_n$ one takes the trace-zero diagonal matrices, so $\mathrm{SL}_n$ has rank $n - 1$; in $\mathfrak{sl}_2$ this is the line $k\,h$ of Example 2.7.4.
2. For $\mathrm{GL}_n$, $W = S_n$ permutes the diagonal entries and $k[\mathfrak{t}_n]^{S_n}$ is generated by the elementary symmetric functions. Under Theorem 2.7.6(3) these correspond to the coefficients of the characteristic polynomial $\det(x - X)$ on $\mathfrak{gl}_n$.
3. The strictly upper triangular matrices $\mathfrak{n}_n$ form a nilpotent subalgebra, but not a Cartan subalgebra: they are normalized by the larger algebra $\mathfrak{b}_n$.

The quotient $\mathfrak{t}/W$ reappears in Section 6 as the base of the Kostant slice: for the dual group, $(\mathfrak{g}^\vee)^* /\!\!/ G^\vee \cong (\mathfrak{t}^\vee)^*/W$.

**The dictionary in characteristic zero.** When $\operatorname{char} k = 0$ the Lie algebra captures much of the structure of a connected group:

- connected algebraic subgroups $H \subset G$ are determined by $\mathrm{Lie}(H) \subset \mathfrak{g}$, and $H$ is normal iff $\mathrm{Lie}(H)$ is an ideal;
- $G$ is commutative, solvable or nilpotent iff $\mathfrak{g}$ is commutative, solvable or nilpotent, respectively;

- for $G \subset \mathrm{GL}_n$, $G$ is unipotent iff every element of $\mathfrak{g}$ is a nilpotent matrix (a torus has commutative, hence nilpotent, Lie algebra, but is not unipotent);
- if $G$ is reductive, then $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$ with $\mathfrak{z}$ the centre and $[\mathfrak{g}, \mathfrak{g}]$ semisimple; for $\mathrm{GL}_n$ this is $\mathfrak{gl}_n = k \cdot 1 \oplus \mathfrak{sl}_n$;
- for unipotent $U \subset \mathrm{GL}_n$, the exponential $X \mapsto \sum_{m \geq 0} X^m/m!$ (a finite sum) is an isomorphism of varieties $\mathrm{Lie}(U) \to U$.

In positive characteristic several of these fail; for instance, the Frobenius $\mathbb{G}_a \to \mathbb{G}_a$, $x \mapsto x^p$, is a bijective homomorphism with zero differential.

# 3 Root data and the Langlands dual group

## 3.1 Root data

Let $G$ be connected reductive and $T \subset G$ a maximal torus. Write

$$X^* = X^*(T) = \mathrm{Hom}(T, \mathbb{G}_m), \qquad X_* = X_*(T) = \mathrm{Hom}(\mathbb{G}_m, T)$$

for the character and cocharacter lattices. Composition gives a perfect pairing

$$\langle\ ,\ \rangle : X^* \times X_* \to \mathbb{Z}, \qquad (\chi \circ \lambda)(s) = s^{\langle \chi, \lambda \rangle},$$

since $\mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) = \mathbb{Z}$.

The torus $T$ acts on $\mathfrak{g} = \mathrm{Lie}(G)$ through the adjoint representation, and

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha, \qquad \mathfrak{t} = \mathrm{Lie}(T),$$

where $\mathfrak{t}$ is the Cartan subalgebra of Theorem 2.7.6 and $\Phi \subset X^* \setminus \{0\}$ is the set of *roots*. Each $\mathfrak{g}_\alpha$ is one-dimensional. For each root $\alpha$ there is a homomorphism $\varphi_\alpha : \mathrm{SL}_2 \to G$, unique up to conjugation by $T$, whose differential maps the upper and lower triangular nilpotent matrices to $\mathfrak{g}_\alpha$ and $\mathfrak{g}_{-\alpha}$. The *coroot* is

$$\alpha^\vee \in X_*, \qquad \alpha^\vee(s) = \varphi_\alpha \begin{pmatrix} s & 0 \\ 0 & s^{-1} \end{pmatrix},$$

and satisfies $\langle \alpha, \alpha^\vee \rangle = 2$.

**Definition 3.1.1.** Let $V$ be a finite-dimensional vector space over a field of characteristic 0 (or a free abelian group of finite rank). A *reflection* of $V$ is a linear automorphism $s : V \to V$ such that $s \neq 1$, $s^2 = 1$, and $s$ fixes a hyperplane pointwise, i.e. $\ker(s - 1)$ has codimension 1. Equivalently, there are $\alpha \in V$ and $\alpha^\vee \in V^*$ with $\langle \alpha, \alpha^\vee \rangle = 2$ such that

$$s(x) = x - \langle x, \alpha^\vee \rangle\, \alpha \qquad (x \in V).$$

Then $s(\alpha) = -\alpha$, and $s$ fixes the hyperplane $\{x : \langle x, \alpha^\vee \rangle = 0\}$ pointwise. We write $s = s_{\alpha, \alpha^\vee}$, or $s_\alpha$ when $\alpha^\vee$ is understood.

**Example 3.1.2.** On $V = \mathbb{Z}^n$, take $\alpha = e_i - e_j$ and $\alpha^\vee = e_i^\vee - e_j^\vee$. Then $s_\alpha(x) = x - (x_i - x_j)(e_i - e_j)$ swaps the $i$-th and $j$-th coordinates and fixes the hyperplane $x_i = x_j$. On $\mathbb{Z}^2$ with $\alpha = (1, -1)$ this is the reflection across the diagonal line $x_1 = x_2$.

For each root $\alpha$ of $G$, the pair $(\alpha, \alpha^\vee)$ defines reflections $s_\alpha$ of $X^*$ and, by the same formula with roles exchanged, $s_{\alpha^\vee}(\lambda) = \lambda - \langle \alpha, \lambda \rangle \alpha^\vee$ of $X_*$. The reflections $s_\alpha$ preserve $\Phi$, and the Weyl group $W = N_G(T)/T$ of Definition 2.3.6 is generated by them: the image of $\varphi_\alpha \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ in $W$ acts on $X^*$ as $s_\alpha$. For $\mathrm{GL}_n$, the reflection in the root $e_i - e_j$ is the transposition $(i\,j) \in S_n$.

**Definition 3.1.3.** A *root datum* is a quadruple $(X, \Phi, X^\vee, \Phi^\vee)$ where $X, X^\vee$ are free abelian groups of finite rank in perfect duality, $\Phi \subset X$ and $\Phi^\vee \subset X^\vee$ are finite subsets with a bijection $\alpha \mapsto \alpha^\vee$, such that $\langle \alpha, \alpha^\vee \rangle = 2$ and the reflections $s_\alpha$, $s_{\alpha^\vee}$ preserve $\Phi$, $\Phi^\vee$ respectively. It is *reduced* if $2\alpha \notin \Phi$ for $\alpha \in \Phi$.

The *root datum of* $G$ is $\Psi(G) = (X^*, \Phi, X_*, \Phi^\vee)$; it is independent of $T$ up to isomorphism by Theorem 2.3.5.

**Theorem 3.1.4** (Chevalley; Demazure–Grothendieck)**.** *The assignment $G \mapsto \Psi(G)$ induces a bijection between isomorphism classes of connected reductive groups over $k$ (equivalently, split reductive groups over any field) and isomorphism classes of reduced root data.*

See [Mil17, Ch. 21, 23 and App. C].

## 3.2 The dual group

**Definition 3.2.1.** The *dual* of a root datum $\Psi = (X, \Phi, X^\vee, \Phi^\vee)$ is $\Psi^\vee = (X^\vee, \Phi^\vee, X, \Phi)$. The *Langlands dual group* of $G$ is the connected reductive group $G^\vee$ over $\mathbb{C}$ with

$$\Psi(G^\vee) = \Psi(G)^\vee.$$

It exists and is unique up to isomorphism by Theorem 3.1.4. Some immediate consequences:

- The maximal torus of $G^\vee$ is $T^\vee = X^*(T) \otimes \mathbb{C}^\times$, so $X^*(T^\vee) = X_*(T)$.
- $G$ and $G^\vee$ have isomorphic Weyl groups.
- $G^\vee$ depends only on the combinatorics of $G$; it is a complex group even when $G$ is defined over $\mathbb{F}_q$ or $\mathbb{Q}_p$.
- The centre $Z(G^\vee)$ is Cartier dual to the algebraic fundamental group $\pi_1(G) = X_*/\mathbb{Z}\Phi^\vee$. In particular a semisimple group $G$ is simply connected iff $G^\vee$ is adjoint.

**Example 3.2.2** ($\mathrm{GL}_n$ is self-dual)**.** Let $T = \{\mathrm{diag}(t_1, \dots, t_n)\}$. Then $X^* = \mathbb{Z}^n$ with basis $e_i(t) = t_i$, and $X_* = \mathbb{Z}^n$ with basis

$$e_i^\vee(s) = \mathrm{diag}(1, \dots, \underset{i}{s}, \dots, 1), \qquad \langle e_i, e_j^\vee \rangle = \delta_{ij}.$$

The torus acts on the matrix unit $E_{ij}$ by $t_i t_j^{-1}$, so $\Phi = \{e_i - e_j : i \neq j\}$. The copy of $\mathrm{SL}_2$ in rows and columns $i, j$ gives

$$\alpha_{ij}^\vee(s) = \mathrm{diag}(\dots, \underset{i}{s}, \dots, \underset{j}{s^{-1}}, \dots), \qquad \alpha_{ij}^\vee = e_i^\vee - e_j^\vee,$$

and $\langle e_i - e_j, e_i^\vee - e_j^\vee \rangle = 2$. The dual datum $(\mathbb{Z}^n, \{e_i^\vee - e_j^\vee\}, \mathbb{Z}^n, \{e_i - e_j\})$ is the same datum with letters renamed, so

$$\mathrm{GL}_n^\vee \cong \mathrm{GL}_n(\mathbb{C}).$$

**Example 3.2.3** ($\mathrm{SL}_n$ and $\mathrm{PGL}_n$)**.** Both groups have roots $e_i - e_j$; only the lattices change. With $\mathbf{1} = (1, \dots, 1)$:

| | $X^*$ | $X_*$ |
|---|---|---|
| $\mathrm{SL}_n$ | $\mathbb{Z}^n/\mathbb{Z}\mathbf{1}$ | $\{a \in \mathbb{Z}^n : \sum a_i = 0\}$ |
| $\mathrm{PGL}_n$ | $\{a \in \mathbb{Z}^n : \sum a_i = 0\}$ | $\mathbb{Z}^n/\mathbb{Z}\mathbf{1}$ |

The two rows are exchanged by duality, so $\mathrm{SL}_n^\vee \cong \mathrm{PGL}_n(\mathbb{C})$ and $\mathrm{PGL}_n^\vee \cong \mathrm{SL}_n(\mathbb{C})$. Consistently with the remark on centres, $Z(\mathrm{SL}_n) = \mu_n$ is dual to $\pi_1(\mathrm{PGL}_n) = \mathbb{Z}/n$, and $\pi_1(\mathrm{SL}_n) = 0$ is dual to $Z(\mathrm{PGL}_n) = 1$. For $n = 2$: $\mathrm{SL}_2$ has $X^* = \mathbb{Z}$ with root 2 and $X_* = \mathbb{Z}$ with coroot 1; swapping gives root 1 and coroot 2, the root datum of $\mathrm{PGL}_2$.

**Example 3.2.4** (Types $B$ and $C$)**.** For $\mathrm{SO}_5$ (type $B_2$), $\Phi = \{\pm e_1, \pm e_2\} \cup \{\pm e_1 \pm e_2\}$, the first set short and the second long. The coroots are $\Phi^\vee = \{\pm 2e_1^\vee, \pm 2e_2^\vee\} \cup \{\pm e_1^\vee \pm e_2^\vee\}$, now with the first set long: this is type $C_2$. In general $\mathrm{SO}_{2n+1}^\vee \cong \mathrm{Sp}_{2n}(\mathbb{C})$ and $\mathrm{Sp}_{2n}^\vee \cong \mathrm{SO}_{2n+1}(\mathbb{C})$.

| $G$ | $G^\vee$ | $G$ | $G^\vee$ |
|---|---|---|---|
| $\mathrm{GL}_n$ | $\mathrm{GL}_n$ | $\mathrm{SO}_{2n}$ | $\mathrm{SO}_{2n}$ |
| $\mathrm{SL}_n$ | $\mathrm{PGL}_n$ | $\mathrm{Spin}_{2n+1}$ | $\mathrm{PSp}_{2n}$ |
| $\mathrm{SO}_{2n+1}$ | $\mathrm{Sp}_{2n}$ | $G_2, F_4, E_8$ | $G_2, F_4, E_8$ |
| $\mathbb{G}_m$ | $\mathbb{G}_m$ | | |

### 3.3 Why the dual group appears

**Theorem 3.3.1** (Satake)**.** *Let $G$ be split reductive over $\mathbb{Q}_p$. Unramified irreducible representations of $G(\mathbb{Q}_p)$ are in bijection with semisimple conjugacy classes in $G^\vee(\mathbb{C})$.*

If $s_\pi$ is the class attached to $\pi$ (its *Satake parameter*) and $V$ is a representation of $G^\vee$, the local $L$-factor is

$$L(s, \pi, V) = \det\left(1 - s_\pi\, p^{-s} \mid V\right)^{-1}.$$

For $G = \mathrm{GL}_n$, $s_\pi = \mathrm{diag}(\alpha_1, \dots, \alpha_n)$ and the standard representation gives $L(s,\pi) = \prod_i (1 - \alpha_i p^{-s})^{-1}$. For $n = 1$ the Satake bijection is local class field theory for unramified characters. See [Bor79, Gro98].

The geometric Satake equivalence of Mirković–Vilonen gives an intrinsic construction: $G^\vee$ is the Tannakian group of the category of $G(\mathcal{O})$-equivariant perverse sheaves on the affine Grassmannian $\mathrm{Gr}_G = G(F)/G(\mathcal{O})$, $F = \mathbb{C}((t))$ [Zhu17]. Relative Langlands duality may be viewed as asking what replaces $G^\vee$ when $\mathrm{Gr}_G$ is replaced by $X(F)/G(\mathcal{O})$ for a $G$-variety $X$. The answer will be a Hamiltonian $G^\vee$-space.

## 4 Periods and spherical varieties

### 4.1 Periods

Let $F$ be a global field with adeles $\mathbb{A}$, and write $[G] = G(F)\backslash G(\mathbb{A})$. For an algebraic subgroup $H \subset G$ and an automorphic form $\varphi$ on $[G]$, the *$H$-period* of $\varphi$ is

$$P_H(\varphi) = \int_{[H]} \varphi(h)\, dh,$$

when convergent. More generally, a $G$-variety $X$ defines a theta series on $[G]$ from Schwartz functions on $X(\mathbb{A})$, and the $X$-period is the pairing of $\varphi$ with it; for $X = H\backslash G$ one recovers $P_H$.

**Example 4.1.1** (Hecke)**.** Let $G = \mathrm{PGL}_2$, $A = \{\mathrm{diag}(a, 1)\} \cong \mathbb{G}_m$ and $\varphi$ in a cuspidal automorphic representation $\pi$. Then

$$\int_{F^\times\backslash\mathbb{A}^\times} \varphi\begin{pmatrix} a & \\ & 1 \end{pmatrix} |a|^{s-\frac{1}{2}}\, d^\times a \;\doteq\; L(s, \pi).$$

At $s = \frac{1}{2}$, the $A$-period of $\varphi$ is the central value $L(\frac{1}{2}, \pi)$, the $L$-function of the standard representation $\mathbb{C}^2$ of $G^\vee = \mathrm{SL}_2(\mathbb{C})$.

Thus the automorphic datum $(\mathrm{PGL}_2,\ A\backslash\mathrm{PGL}_2)$ corresponds to the spectral datum $(\mathrm{SL}_2,\ \mathbb{C}^2)$. This is the first example of a dual pair.

### 4.2 Spherical varieties

**Definition 4.2.1.** A normal $G$-variety $X$ is *spherical* if a Borel subgroup of $G$ has an open orbit on $X$.

**Example 4.2.2.**

1. Toric varieties, for $G$ a torus.
2. Flag varieties $G/P$, by the Bruhat decomposition.
3. The *group case*: $X = G$ under $G \times G$ acting by $(g_1, g_2) \cdot x = g_1 x g_2^{-1}$. The big Bruhat cell is an open orbit of $B \times B^-$.
4. Symmetric varieties $G/G^\theta$ for an involution $\theta$, e.g. $\mathrm{GL}_{2n}/\mathrm{Sp}_{2n}$ and $\mathrm{GL}_{2n}/(\mathrm{GL}_n \times \mathrm{GL}_n)$.
5. $A\backslash \mathrm{PGL}_2$ from Example 4.1.1, and $(\mathrm{GL}_{n+1} \times \mathrm{GL}_n)/\Delta\mathrm{GL}_n$, which underlies Rankin–Selberg and Gan–Gross–Prasad periods.

The relevance of sphericity is the *multiplicity one* phenomenon: for a local field $F$ and many spherical $X$,

$$\dim \mathrm{Hom}_{G(F)}\left(\pi, C^\infty(X(F))\right) \leq 1$$

for every irreducible $\pi$. When local multiplicities are at most one, a global period is forced to factor as a product of local functionals, and an Euler product, hence an $L$-function, appears. Multiplicity one does not hold for every spherical variety, and the definition of hyperspherical variety in Section 5 is in part an abstraction of this property [BZSV24, Rem. 1.2.2].

**Example 4.2.3.** In the group case, $\mathrm{Hom}_{G\times G}(\pi \boxtimes \sigma, C^\infty(G))$ is nonzero iff $\sigma \cong \pi^\vee$, and is then one-dimensional by Schur's lemma. The corresponding global period is the Petersson inner product, related to $L(1, \pi, \mathrm{Ad})$.

See [Pez10, Bri12] for introductions to spherical varieties.

## 5 Hamiltonian spaces and hyperspherical varieties

A basic insight of [BZSV24] is that periods are *microlocal*: they should be attached not to $X$ but to $T^*X$, viewed as a Hamiltonian $G$-space. Symmetries such as the Fourier transform underlying Tate's thesis are visible on $T^*\mathbb{A}^1$ but not on $\mathbb{A}^1$. Moreover, important periods, such as Whittaker periods and theta correspondences, come from Hamiltonian spaces that are not cotangent bundles at all. For background on the symplectic geometry below see [CG97, Ch. 1]; for a detailed treatment of hyperspherical varieties see [BZSV24, §3] and the notes [Ree23].

### 5.1 Algebraic symplectic geometry

Let $M$ be a smooth variety over $\mathbb{C}$ and $\Omega^p_M = \wedge^p \Omega^1_M$ the sheaf of Kähler $p$-forms. A *symplectic form* is $\omega \in H^0(M, \Omega^2_M)$ with $d\omega = 0$ such that $T_M \to \Omega^1_M$, $v \mapsto \iota_v \omega$, is an isomorphism.

**Example 5.1.1.** On $T^*\mathbb{A}^n = \mathrm{Spec}\,\mathbb{C}[q_1, \ldots, q_n, p_1, \ldots, p_n]$, $\omega = \sum dp_i \wedge dq_i$. More generally, for smooth $X$ the cotangent bundle $T^*X$ carries the canonical 1-form $\theta$, locally $\sum p_i\, dq_i$, and $\omega = d\theta$ is symplectic.

Each $f \in \mathcal{O}(M)$ has a *Hamiltonian vector field* $H_f$ defined by $\iota_{H_f}\omega = df$, and $\{f, g\} = \omega(H_f, H_g)$ is a Poisson bracket on $\mathcal{O}(M)$.

Let $\mathfrak{g}^* = \mathrm{Hom}(\mathfrak{g}, \mathbb{C})$ be the dual of the Lie algebra, with the coadjoint action $(g \cdot \lambda)(\xi) = \lambda(\mathrm{Ad}(g^{-1})\xi)$.

**Definition 5.1.2.** A *Hamiltonian $G$-space* is a symplectic variety $M$ with a $G$-action preserving $\omega$ and a $G$-equivariant *moment map* $\mu : M \to \mathfrak{g}^*$ such that, for every $\xi \in \mathfrak{g}$, the vector field generated by $\xi$ is $H_{\mu_\xi}$, where $\mu_\xi = \langle \mu, \xi \rangle$.

**Example 5.1.3.** If $G$ acts on a smooth variety $X$, then $T^*X$ is a Hamiltonian $G$-space with

$$\mu(x, p)(\xi) = p(\xi \cdot x),$$

where $\xi \cdot x \in T_x X$ is the infinitesimal action. For $G \times G$ acting on $T^*G \cong G \times \mathfrak{g}^*$ (trivialized by left translation), $\mu(g, \lambda) = (\mathrm{Ad}^*_g \lambda, -\lambda)$.

**Definition 5.1.4.** The *Hamiltonian reduction* of $M$ by a subgroup $H$ is $M/\!\!/\!\!/H = \mu_H^{-1}(0)/H$, where $\mathfrak{h} = \mathrm{Lie}(H)$ and $\mu_H$ is the composite $M \to \mathfrak{g}^* \to \mathfrak{h}^*$ of $\mu$ with the restriction map dual to the inclusion $\mathfrak{h} \subset \mathfrak{g}$.

For instance $T^*G/\!\!/\!\!/H \cong T^*(G/H)$, with $H$ acting by right translation.

**Exercise 5.1.5.** Verify that the map in Example 5.1.3 is a moment map.

## 5.2 The grading

**Definition 5.2.1.** A *graded Hamiltonian $G$-space* is a Hamiltonian $G$-space $M$ with an action of $\mathbb{G}_{\mathrm{gr}} \cong \mathbb{G}_m$ commuting with $G$ such that $a^*\omega = a^2\omega$ for $a \in \mathbb{G}_{\mathrm{gr}}$. The moment map is then $\mathbb{G}_{\mathrm{gr}}$-equivariant for the action $a \cdot \lambda = a^2\lambda$ on $\mathfrak{g}^*$.

On $T^*X$ one may let $\mathbb{G}_{\mathrm{gr}}$ act on cotangent fibres by $p \mapsto a^2 p$, so that $\omega = \sum dp_i \wedge dq_i$ has weight 2. On the spectral side the grading becomes the cohomological grading in the geometric setting and the Frobenius weight in the arithmetic setting; see [BZSV24, §1.4, §2.8] for the precise conventions.

## 5.3 Hyperspherical varieties

**Definition 5.3.1** ([BZSV24, §3.5.1])**.** A graded Hamiltonian $G$-space $M$ (smooth and irreducible) is *hyperspherical* if

1. $M$ is affine;
2. $M$ is *coisotropic*: the field $\mathbb{C}(M)^G$ of $G$-invariant rational functions is commutative for the Poisson bracket; equivalently, generic $G$-orbits are coisotropic;
3. the image of the moment map meets the nilpotent cone of $\mathfrak{g}^*$;
4. the stabilizer in $G$ of a generic point of $M$ is connected;
5. the $\mathbb{G}_{\mathrm{gr}}$-action is *neutral*, a normalization condition on the grading defined in [BZSV24, §3.5.4].

Condition (2) is the Hamiltonian counterpart of sphericity: by work of Knop, a smooth $G$-variety $X$ is spherical if and only if $T^*X$ is coisotropic. Conditions (3)–(5) are what make the structure theorem below work.

## 5.4 The structure theorem

**Theorem 5.4.1** ([BZSV24, Thm. 3.6.1])**.** *Every hyperspherical $G$-space is obtained by* Whittaker induction *from a triple*

$$\big(H \subset G \text{ reductive}, \quad \iota : \mathrm{SL}_2 \to G \text{ commuting with } H, \quad S \text{ a symplectic representation of } H\big).$$

More precisely [BZSV24, Thm. 3.6.1], $H$ is the stabilizer of a point $x$ in the closed $G\times\mathbb{G}_{\mathrm{gr}}$-orbit of $M$, and $S$ is the fibre at $x$ of the symplectic normal bundle to that orbit. When $\iota$ is trivial, Whittaker induction is ordinary Hamiltonian induction [BZSV24, §3.3]:

$$M = \big(T^*G \times S\big)/\!\!/\!\!/ H \cong G \times^H \big(S \times_{\mathfrak{h}^*} \mathfrak{g}^*\big) \cong G \times^H \big(\mathfrak{h}^\perp \times S\big), \qquad \mathfrak{h}^\perp = (\mathfrak{g}/\mathfrak{h})^* \subset \mathfrak{g}^*,$$

where $S \to \mathfrak{h}^*$ is the moment map of $S$, and the last isomorphism uses an $H$-equivariant splitting of $\mathfrak{g}^* \to \mathfrak{h}^*$. Here $G\times^H Y = (G\times Y)/H$ denotes the quotient by the action $h\cdot(g,y) = (gh^{-1}, hy)$. With $S = 0$ this recovers $T^*(G/H)$. A nontrivial $\iota$ twists the construction by a Whittaker datum along the nilpotent element $\iota\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)$.

*Remark* 5.4.2. The triple in Theorem 5.4.1 matches the classification of half-BPS boundary conditions of $\mathcal{N}=4$ super Yang–Mills theory by Gaiotto and Witten [GW09]: a Nahm pole ($\iota$), a subgroup ($H$), and a theory of hypermultiplets ($S$). See [BZSV24, Rem. 1.2.1] and Section 7.

### 5.5 The duality

BZSV expect [BZSV24, §5.2] that for "anomaly-free" hyperspherical $(G, M)$ there is a dual hyperspherical $(G^\vee, M^\vee)$, with $(M^\vee)^\vee \cong M$. For polarized $M = T^*X$ they construct a candidate $M^\vee$ explicitly [BZSV24, §4] from three ingredients: the dual group $G_X^\vee \subset G^\vee$ of the spherical variety $X$ (Gaitsgory–Nadler, Sakellaridis–Venkatesh, Knop–Schalke), a homomorphism $\mathrm{SL}_2 \to G^\vee$ commuting with $G_X^\vee$, and a symplectic representation $S_X$ of $G_X^\vee$.

| Name | $(G, M)$ | $(G^\vee, M^\vee)$ |
|---|---|---|
| Group case | $(G\times G,\ T^*G)$ | $(G^\vee\times G^\vee,\ T^*G^\vee)$ |
| Whittaker–trivial | $(G,\ \mathrm{pt})$ | $(G^\vee,\ T^*_\psi(G^\vee/U^\vee))$ |
| Hecke | $(\mathrm{PGL}_2,\ T^*(A\backslash\mathrm{PGL}_2))$ | $(\mathrm{SL}_2,\ T^*\mathbb{C}^2)$ |
| Iwasawa–Tate | $(\mathbb{G}_m,\ T^*\mathbb{A}^1)$ | $(\mathbb{G}_m,\ T^*\mathbb{A}^1)$ |
| GJ–RS * | $(\mathrm{GL}_n\times\mathrm{GL}_n,\ T^*\mathrm{Mat}_{n\times n})$ | $(\mathrm{GL}_n\times\mathrm{GL}_n,\ T^*(G\times^{\Delta\mathrm{GL}_n}\mathbb{A}^n))$ |
| Gan–Gross–Prasad † | $(\mathrm{SO}_{2n}\times\mathrm{SO}_{2n+1},\ T^*(\Delta\mathrm{SO}_{2n}\backslash G))$ | $(\mathrm{SO}_{2n}\times\mathrm{Sp}_{2n},\ \mathrm{std}\otimes\mathrm{std})$ |

*Godement–Jacquet and Rankin–Selberg; $G = \mathrm{GL}_n\times\mathrm{GL}_n$. †$G = \mathrm{SO}_{2n}\times\mathrm{SO}_{2n+1}$, with $\mathrm{SO}_{2n}$ embedded diagonally.

A selection from [BZSV24, Table 1.5.1]. The Hecke row recovers Example 4.1.1: $M^\vee = T^*\mathbb{C}^2$ is the cotangent bundle of the standard representation of $\mathrm{SL}_2$. The Whittaker row, read from right to left, explains the special role of the Whittaker model: the Whittaker period for a group is dual to the point for its dual group, i.e. to the trivial $L$-function. In the Gan–Gross–Prasad row, $M^\vee$ is a symplectic representation that is not a cotangent bundle; such cases correspond to periods whose square, rather than the period itself, is an $L$-value.

## 6 The conjectures

Each conjecture asserts that an object built from $(G, M)$ on the automorphic side matches an object built from $(G^\vee, M^\vee)$ on the spectral side.

### 6.1 Shearing

For a $\mathbb{G}_m$-equivariant complex $N = \bigoplus_i N_i$, with $N_i$ the weight-$i$ part, the *shear* is $N^\natural = \bigoplus_i N_i[i]$. Applied to $\mathcal{O}(M^\vee)$ with its $\mathbb{G}_{\mathrm{gr}}$-grading, shearing converts the grading into a cohomological shift. For example, linear functions on $\mathfrak{g}^*$ have $\mathbb{G}_{\mathrm{gr}}$-weight $-2$ (since $\mathbb{G}_{\mathrm{gr}}$ scales $\mathfrak{g}^*$

by $a^2$), so $\mathcal{O}(\mathfrak{g}^*) = \mathrm{Sym}(\mathfrak{g})$ becomes $\mathrm{Sym}(\mathfrak{g}[-2])$, with generators in cohomological degree 2, and $\big(\mathcal{O}(\mathfrak{g}^*)^G\big)^\natural \cong H^*(BG)$. See [BZSV24, §6].

### 6.2 The local conjecture

Let $X$ be a smooth affine spherical $G$-variety such that $M = T^*X$ is hyperspherical, with dual $M^\vee$, and let $F = \mathbb{C}((t))$ and $\mathcal{O} = \mathbb{C}[[t]]$.

**Conjecture 6.2.1** ([BZSV24, §7.5])**.** *There is an equivalence of categories, compatible with the Hecke actions through derived geometric Satake,*

$$\mathrm{Shv}\big(X(F)/G(\mathcal{O})\big) \simeq \mathrm{QC}^\natural\big(M^\vee/G^\vee\big).$$

Two consistency checks, each valid up to renormalization and support conditions:

1. For the group case $X = G$ under $G \times G$, the left side is $\mathrm{Shv}_{G(\mathcal{O})}(\mathrm{Gr}_G)$ and $M^\vee/(G^\vee \times G^\vee) \cong (\mathfrak{g}^\vee)^*/G^\vee$. Conjecture 6.2.1 becomes the derived geometric Satake equivalence of Bezrukavnikov–Finkelberg.
2. For $X = \mathrm{pt}$ the left side is $\mathrm{Shv}(BG(\mathcal{O}))$, controlled by $H^*(BG)$. The dual is the Whittaker space, and $M^\vee/G^\vee$ is identified with the Kostant slice $\cong (\mathfrak{g}^\vee)^*/\!\!/G^\vee$, whose sheared ring of functions is $H^*(BG)$.

### 6.3 The global conjecture

Let $\Sigma$ be a smooth projective curve over $\mathbb{C}$. On the automorphic side, let

$$\mathrm{Bun}_G^X = \{(P, s) : P \text{ a } G\text{-bundle on } \Sigma,\ s \text{ a section of } P \times^G X\},$$

with projection $\pi : \mathrm{Bun}_G^X \to \mathrm{Bun}_G$, and define the *period sheaf* $\mathcal{P}_X = \pi_! \underline{k} \in \mathrm{Shv}(\mathrm{Bun}_G)$. On the spectral side, let

$$\mathrm{Loc}_{G^\vee}^{X^\vee} = \{(E, s) : E \text{ a } G^\vee\text{-local system},\ s \text{ a flat section of } E \times^{G^\vee} X^\vee\},$$

with projection $p : \mathrm{Loc}_{G^\vee}^{X^\vee} \to \mathrm{Loc}_{G^\vee}$, and define the *L-sheaf* $\mathcal{L}_{X^\vee} = p_*\,\omega \in \mathrm{IndCoh}(\mathrm{Loc}_{G^\vee})$, the pushforward of the dualizing sheaf.

**Conjecture 6.3.1** ([BZSV24, §12.1])**.** *Under the geometric Langlands equivalence*

$$\mathrm{Shv}(\mathrm{Bun}_G) \simeq \mathrm{IndCoh}_{\mathcal{N}}(\mathrm{Loc}_{G^\vee})$$

*(in a suitable sheaf-theoretic setting, e.g. D-modules on* $\mathrm{Bun}_G$*), the normalized period sheaf* $\mathcal{P}_X$ *corresponds to the normalized L-sheaf* $\mathcal{L}_{X^\vee}$*.*

The name is explained by the stalks: at a local system $E$, the stalk of $\mathcal{L}_{X^\vee}$ is computed by the space of flat sections of $E \times^{G^\vee} X^\vee$; for $X^\vee = V$ a vector space this involves $H^*(\Sigma, E_V)$, whose Frobenius trace in the arithmetic setting is an $L$-value. Feng and Wang have studied the Hecke case.

### 6.4 The numerical conjecture

**Conjecture 6.4.1** ([BZSV24, §14.2], schematic)**.** *Let $F$ be a global function field and $\pi$ an everywhere unramified tempered cuspidal automorphic representation of $G$ with Langlands parameter $\phi_\pi$. For $\varphi \in \pi$, the normalized period satisfies*

$$\frac{|P_X(\varphi)|^2}{\langle \varphi, \varphi \rangle} \doteq \frac{L_{M^\vee}(\phi_\pi)}{L(1, \pi, \mathrm{Ad})},$$

*where $L_{M^\vee}$ is an L-value attached to $M^\vee$. If $M^\vee = S$ is a symplectic representation of $G^\vee$ with $\mathbb{G}_{\mathrm{gr}}$ acting by scaling, then $L_{M^\vee}(\phi_\pi) = L(\frac{1}{2}, \pi, S)$. If moreover $S = T^*V = V \oplus V^*$ is polarized, then $P_X(\varphi) \doteq L(\frac{1}{2}, \pi, V)$ for suitably normalized $\varphi$.*

For the Hecke period this reads $|P_A(\varphi)|^2/\langle\varphi,\varphi\rangle \doteq L(\frac{1}{2},\pi)^2/L(1,\pi,\mathrm{Ad})$. The conjecture is modelled on classical results (Hecke, Iwasawa–Tate, Rankin–Selberg, Whittaker coefficients) and on the Ichino–Ikeda conjecture for Gan–Gross–Prasad periods, where the right-hand side is $L(\frac{1}{2}, \pi_1 \times \pi_2)/L(1,\pi,\mathrm{Ad})$. See [BZSV24, §14.5–14.6].

# 7 The field-theoretic picture

## 7.1 Langlands duality as a four-dimensional field theory

A fully extended four-dimensional topological field theory assigns a number to a closed 4-manifold, a vector space to a closed 3-manifold, a category to a closed surface, and a 2-category to a closed 1-manifold. Kapustin and Witten [KW07] argued that a twisted form of $\mathcal{N} = 4$ super Yang–Mills theory gives TFTs $\mathcal{A}_G$ (A-model) and $\mathcal{B}_{G^\vee}$ (B-model), and that electric-magnetic S-duality predicts

$$\mathcal{A}_G \simeq \mathcal{B}_{G^\vee}.$$

Evaluated on a curve $\Sigma$, this is the geometric Langlands equivalence between $\mathrm{Shv}(\mathrm{Bun}_G)$ and (a version of) $\mathrm{QC}(\mathrm{Loc}_{G^\vee})$. Hecke operators are line operators, and their duality is geometric Satake.

In arithmetic topology (Mazur, Kapranov, Reznikov, Morishita, Kim) global fields (number fields and function fields of curves over $\mathbb{F}_q$) behave like 3-manifolds, local fields such as $\mathbb{Q}_p$ and curves over an algebraically closed field like 2-manifolds, and $\mathbb{C}((t))$ like a 1-manifold. BZSV propose to view the whole Langlands program as an equivalence of two "four-dimensional arithmetic field theories", see [BZSV24, §1.3].

## 7.2 Boundary conditions

Gaiotto and Witten [GW09] explained that a Hamiltonian $G$-space $M$ defines a boundary condition for four-dimensional $G$-gauge theory, namely a three-dimensional sigma model into $M$ coupled to the bulk gauge field.

**Conjecture 7.2.1** (Meta-conjecture, [BZSV24, Conj. 1.3.2])**.** *Under $\mathcal{A}_G \simeq \mathcal{B}_{G^\vee}$, the boundary condition $\Theta_M$ for $\mathcal{A}_G$ attached to $M$ corresponds to the boundary condition $\mathcal{L}_{M^\vee}$ for $\mathcal{B}_{G^\vee}$ attached to $M^\vee$, for dual hyperspherical pairs $(G, M) \leftrightarrow (G^\vee, M^\vee)$.*

The conjectures of Section 6 are evaluations of this single statement. Place the theory on $N \times [0,1]$ with $N \times \{0\}$ labelled by $M$; then $N$ acquires a distinguished element of whatever the theory assigns to $N$:

| $N$ | automorphic output | matching statement |
|---|---|---|
| global field ("3d") | period functional on automorphic forms | numerical conjecture |
| curve over $\mathbb{C}$ ("2d") | period sheaf in $\mathrm{Shv}(\mathrm{Bun}_G)$ | global conjecture |
| $\mathbb{C}((t))$ ("1d") | module category $\mathrm{Shv}(X(F)/G(\mathcal{O}))$ | local conjecture |
| interval with two boundaries | a number | relative trace formula |

The field theory picture is not merely organizational: it predicts compatibilities between the conjectures, such as local-global compatibility through $\Theta$-series [BZSV24, §15], and suggests new objects, such as the $L$-sheaf and the Plancherel algebra, which is related to the Coulomb branches of Braverman–Finkelberg–Nakajima [BZSV24, §8].

# 8 Shifted symplectic geometry and open questions

## 8.1 Hamiltonian spaces as Lagrangians

Pantev, Toën, Vaquié and Vezzosi [PTVV13] defined $n$-shifted symplectic structures on derived Artin stacks, and Lagrangian structures on maps into them. The derived intersection of two Lagrangians in an $n$-shifted symplectic stack is $(n-1)$-shifted symplectic. For reductive $G$ with a nondegenerate invariant form on $\mathfrak{g}$, the classifying stack $BG$ is 2-shifted symplectic, and

$$[\mathfrak{g}^*/G] \cong T^*[1]BG$$

is 1-shifted symplectic.

**Theorem 8.1.1** (Calaque [Cal15], Safronov [Saf16])**.** *A Lagrangian structure on $[M/G] \to [\mathfrak{g}^*/G]$ is equivalent to the structure of a Hamiltonian $G$-space on $M$.*

Consequently Hamiltonian reduction is a Lagrangian intersection,

$$[\mu^{-1}(0)/G] \cong [M/G] \times_{[\mathfrak{g}^*/G]} BG,$$

where $\mu^{-1}(0)$ is the derived zero fibre, and is therefore 0-shifted symplectic as a derived stack, even when 0 is not a regular value of $\mu$; see [Saf17a].

## 8.2 Where derived geometry enters

- *Boundary conditions as Lagrangians.* Heuristically, the classical B-model for $G^\vee$ is an AKSZ-type theory with 3-shifted symplectic target $T^*[3]BG^\vee \cong (\mathfrak{g}^\vee)^*[2]/G^\vee$, and boundary conditions are Lagrangians in it. A Hamiltonian $G^\vee$-space gives a Lagrangian $M^\vee/G^\vee \to (\mathfrak{g}^\vee)^*/G^\vee$ in the 1-shifted target, and the $\mathbb{G}_{\mathrm{gr}}$-grading together with shearing moves it into the 3-shifted setting. This is one explanation of why the grading is part of the data.
- *Mapping stacks.* For a curve $\Sigma$ one has $\mathrm{Loc}_{G^\vee} = \mathrm{Map}(\Sigma_{dR}, BG^\vee)$ and $\mathrm{Loc}_{G^\vee}^{X^\vee} = \mathrm{Map}(\Sigma_{dR}, X^\vee/G^\vee)$. By [PTVV13], $\mathrm{Loc}_{G^\vee}$ is 0-shifted symplectic, and the mapping stack $\mathrm{Map}(\Sigma_{dR}, (\mathfrak{g}^\vee)^*/G^\vee)$, which is $T^*[-1]\mathrm{Loc}_{G^\vee}$, is $(-1)$-shifted symplectic; by [Cal15], applying $\mathrm{Map}(\Sigma_{dR}, -)$ to the Lagrangian $M^\vee/G^\vee \to (\mathfrak{g}^\vee)^*/G^\vee$ gives a Lagrangian in the latter. Heuristically, the $L$-sheaf is a quantization of this Lagrangian.
- *Beyond varieties.* Many interesting dual pairs involve $M$ that are singular, non-affine, or "not even a variety (e.g., a stack or derived scheme)" [BZSV24, p. 24]. For example $M = T^*(U\backslash G)$ under $T \times G$, related to Eisenstein series, is not affine.
- *Morphisms of periods.* BZSV suggest [BZSV24, p. 26] that morphisms between periods should be given by quantizations of Lagrangian correspondences between Hamiltonian $G$-spaces, as in unfolding arguments.

## 8.3 Questions

**Question 8.3.1.** Is there a general construction of $M^\vee$ from $M$? BZSV give one in the polarized case. Can the duality be read off from combinatorial invariants, extending the work of Knop and Losev [BZSV24, §1.6]?

**Question 8.3.2.** For which $(G, M)$ is the duality defined? This is governed by an anomaly condition [BZSV24, §5.1].

**Question 8.3.3.** What is the correct derived framework for singular, non-affine and stacky $M$?

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